Tracking Federal Grant Opportunity Volatility: A Relational Database System for Grants.gov Data

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EPPS 6354 Information Management — Spring 2026 — Professor Karl Ho

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Motivation and Research Context

The federal grants landscape has experienced unprecedented volatility following the 2025 change in presidential administration. Executive Order 14332 expanded the federal government's authority to invoke "termination for convenience" clauses and increased political oversight in the grant awarding process. In the months following this shift, federal agencies have restructured, rescinded, delayed, or frozen funding opportunities at a scale that exceeds prior administrative transitions (Clegg, 2025). These changes have introduced substantial uncertainty for organizations that rely on federal grants to plan and sustain long-term programs.

Grants.gov, the centralized federal grant-listing portal, presents funding opportunities as a current-state snapshot, displaying only those opportunities that are currently open or formally closed. When an opportunity is withdrawn or removed, its prior existence—and any intermediate changes to its terms—are effectively erased from the public record. As a result, organizations that applied for funding, or in some cases were notified of potential awards, are left without clear documentation explaining when, how, or why funding opportunities disappeared. Recent analyses show that such disruptions have had measurable downstream effects on nonprofit operations, staffing, and service delivery, particularly for organizations serving low-income and vulnerable populations (Urban Institute, 2025).

Scholars of public-sector data governance argue that as governments increasingly rely on evidence-based and data-driven decision-making, the governance of the underlying data becomes a matter of democratic accountability (OECD, 2022). When federal agencies alter or remove funding information without preserving an auditable historical record, transparency is reduced, citizen trust is weakened, and the capacity for independent policy analysis is constrained (Osakwe, 2025). This project addresses that accountability gap by constructing a longitudinal relational database that captures, historicizes, and surfaces changes in the federal grant opportunity ecosystem over time.

Project Description

The proposed system ingests periodic snapshots of Grants.gov opportunity listings obtained via automated web scraping. A normalized relational schema stores grant metadata, agency information, and funding categories across linked tables, while a dedicated change log records every detected addition, modification, and removal with timestamps. Snapshot-to-snapshot differencing identifies grants that have been newly posted, altered (e.g., deadline extensions, funding changes), or removed entirely. This approach implements a slowly changing dimension (Type 2) strategy, maintaining current records in the opportunities table while preserving full history of the database in the change log.

A Shiny web application will serve as the front-end interface, enabling users to query active opportunities, filter by agency or funding category, review historical changes, and visualize trends in grant availability over time. The database will be implemented in PostgreSQL; the application will connect via the DBI and RPostgres R packages.

Proposed Schema

Table	Column (selected)	Key	Description
agencies	agency_id, agency_name, agency_code	PK: agency_id	Federal agency reference table
opportunities	opportunity_id, agency_id, opportunity_title, post_date, close_date, award_ceiling, award_floor, estimated_total_funding, status	PK: opportunity_id; FK: agency_id → agencies	Core grant opportunity records from Grants.gov listings
categories	category_code, category_name	PK: category_code (natural key)	Funding category lookup; uses Grants.gov type codes
opportunity_ categories	opportunity_id, category_code	PK: (opportunity_id, category_code); FK: both columns	Junction table linking grants to one or more funding categories
snapshots	snapshot_id, snapshot_date, record_count	PK: snapshot_id	Metadata for each Grants.gov extraction run
change_log	change_id, opportunity_id, snapshot_id, change_type, field_changed, old_value, new_value, detected_date	PK: change_id; FK: opportunity_id → opportunities; FK: snapshot_id → snapshots	Longitudinal record of all detected additions, modifications, and removals

Technologies

Database: PostgreSQL • **Scraping:** R (rvest, httr2) • **Application:** R Shiny • **Connectivity:** DBI, RPostgres • **Hosting:** GitHub

AI Use Disclosure

This project made use of generative artificial intelligence tools, specifically OpenAI's GPT models and Anthropic's Claude models, to support the writing and refinement process. These tools were used to assist with literature review, project planning, structural organization, and stylistic editing of draft text. All substantive intellectual contributions—including the selection of the research topic, formulation of the research motivation, database schema design, methodological approach, interpretation of policy context, verification of sources, and final wording—were produced and evaluated by the author.

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